QuantarisAI



DESIGN SYSTEM · VERSION 2 · OCTOBER 2026

AI-native living industrial ecosystem

An AI system that constructs, simulates, observes and evolves an entire industrial ecosystem in front of the person — built on the approved V1 brand, tokens, components and evidence system.

WHAT CHANGED IN V2

V1 stays. A spatial intelligence layer goes on top.

- **Three spatial levels** in real 3D (Three.js): India Industrial Intelligence Map → Living Industrial Ecosystem → Generative Factory Twin.
- **Visible construction** — 11 workflow states stream while the world materialises.
- **Proactive AI discovery** — markers on the object that matters: ✦ ⚠ ◎ ↗ ♻ ? ↻.
- **Signature moment** — waste → recovery → secondary material → new product; the path turns India green only when earned.
- **Ecosystem expansion** — “Model as new industrial opportunity” grows Factory B.
- **What-if simulation** — CURRENT solid, SIMULATED dashed + hatched, recommendation flips announced.
- **A2UI clarification** — agents request approved controls (CapacitySelector…), never generated UI.
- **Evidence everywhere** — every marker and number opens the Evidence Drawer.
- **Added:** 10 components · 5 interactive screens · 15 reference renders · 16 tokens · scene kit · 3 docs.
- **Unchanged:** brand, palette, typography, V1 components, Evidence Drawer, information architecture.

3D shows what is happening · AI explains what could happen · Simulation calculates what happens · Evidence explains why to trust it.

Every colour means something

saffron · opportunity
#F28A1E · dark #F59A3C

PRIMARY brand accent as a FILL: primary button, active stage, selected scenario, selected ...

ashoka · brand navy
#0B1733 · dark #1C3170

Brand navy as a large fill: app icon tile, brand moments, cover slab. Text on it is on-ash...

blue · system / AI
#1E3A8A · dark #8EA8F5

SYSTEM / INTELLIGENCE: links, info states, system data series, network links on maps, AI-a...

green · earned
#117534 · dark #4CC27A

POSITIVE / SUSTAINABLE only: recovered waste, reduced emissions, saved water, viable circu...

amber · assumption
#8A5A00 · dark #E9B949

ASSUMPTION / UNCERTAINTY: assumption tags, dashed assumption bands, attention states. Text...

red · risk
#B42318 · dark #F2786D

ENVIRONMENTAL RISK / FAILURE / CRITICAL CONSTRAINT: risk tags, increases in emissions or w...

surface
#F7F5F0 · dark #0B1220

Workspace ground: warm off-white in light, deep navy in dark (never pure white or black). ...

ink
#101A2E · dark #EDF1F7

Primary text and numbers on surface, surface-raised, surface-sunken and every -soft tint.

flow-water · NEW
#2B7A9E · dark #5BB4D6

Flow grammar: water in, recovered water back (drop packets). The only resource-specific hu...

scene-building · NEW
#FBFAF7 · dark #24324E

3D scenes: building and machine bodies (white-model architecture).

scene-ground · NEW
#E9E5DC · dark #0E1729

3D scenes: the top of every plinth (map, ecosystem, factory floor).

scene-glass · NEW
#33415E · dark #7F93BD

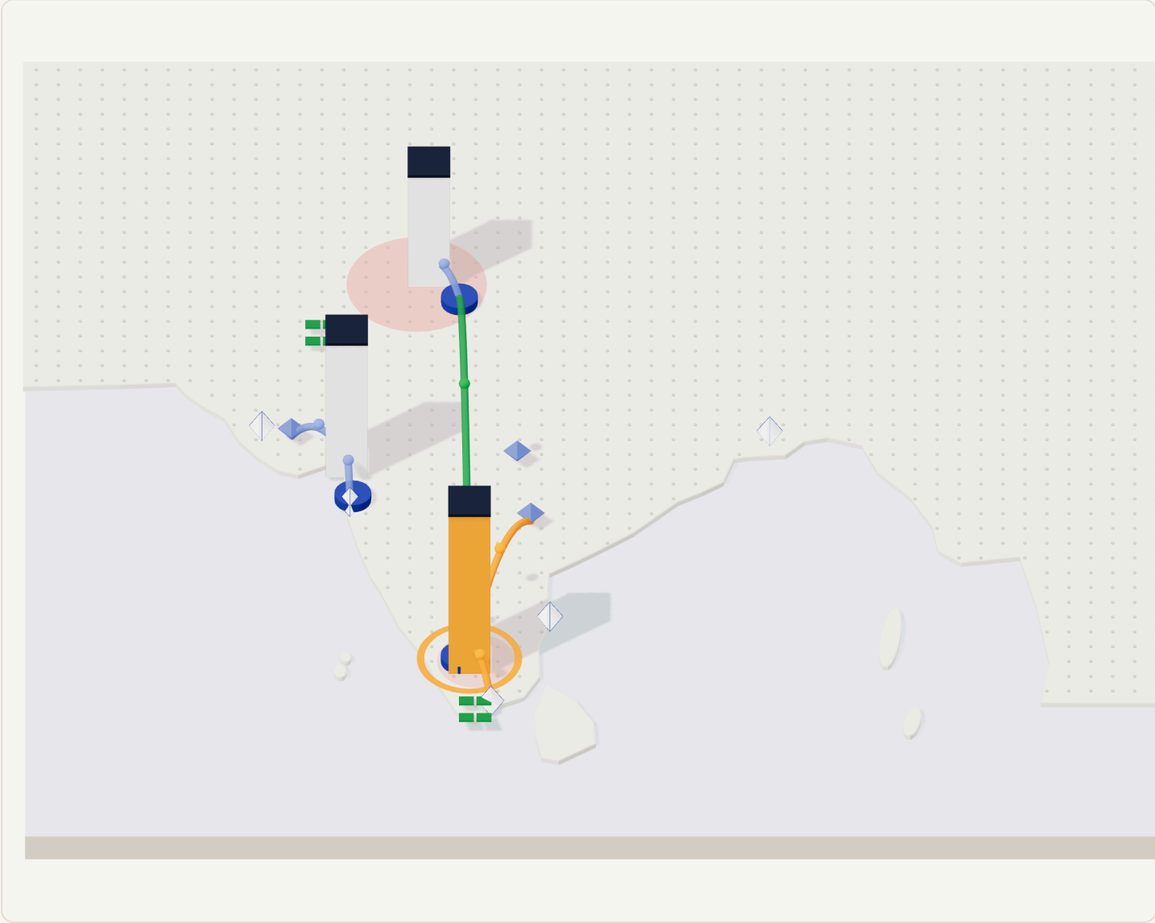
3D scenes: glazing bands, solar panels.

Archivo — display IBM Plex Sans — interface `IBM Plex Mono · 34 t / yr`

LEVEL 1 · WHERE

India Industrial Intelligence Map

A physical map: coastline and islands, no land frontiers drawn. Candidate columns rise by suitability; Tiruppur becomes the saffron recommendation.



Light · supply + export arcs, renewable parks, constraint zones, green circular link



Dark

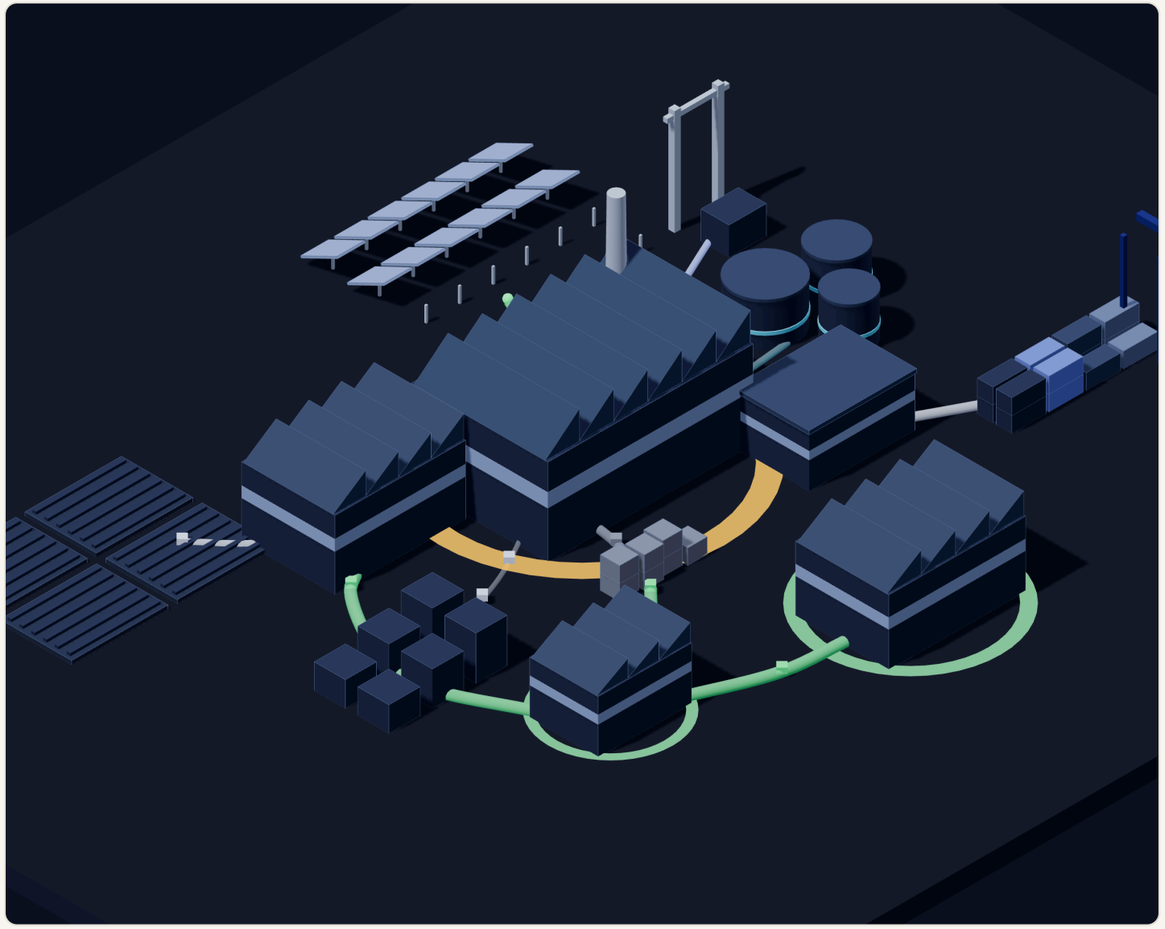
LEVEL 2 · HOW THE SYSTEM INTERACTS

Living Industrial Ecosystem

White-model architecture on a plinth. Colour only where it means something: saffron ring = selected, green rings and flows = earned circularity.



Light · farms, mill, Tiruppur unit, solar + grid, water, workforce, port, waste, recycler, Factory B

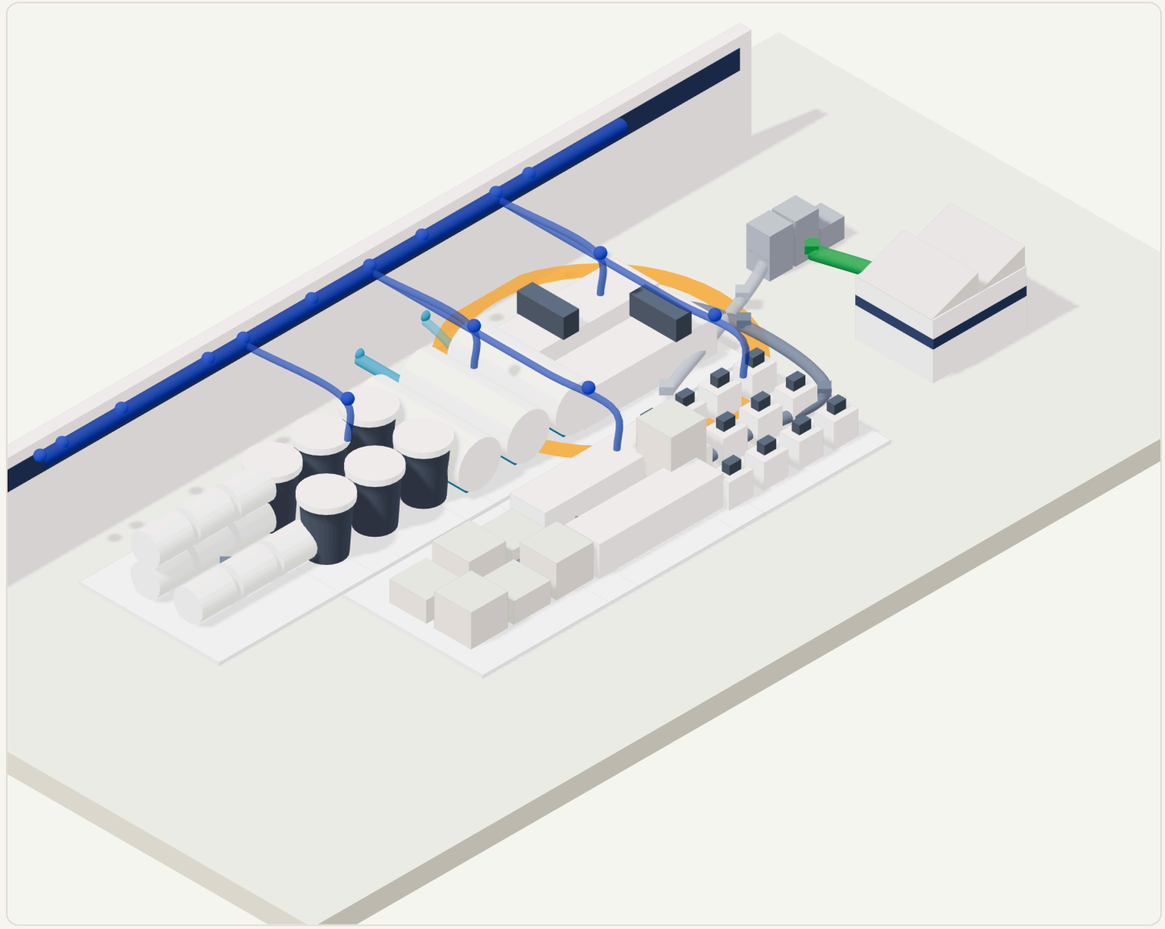


Dark

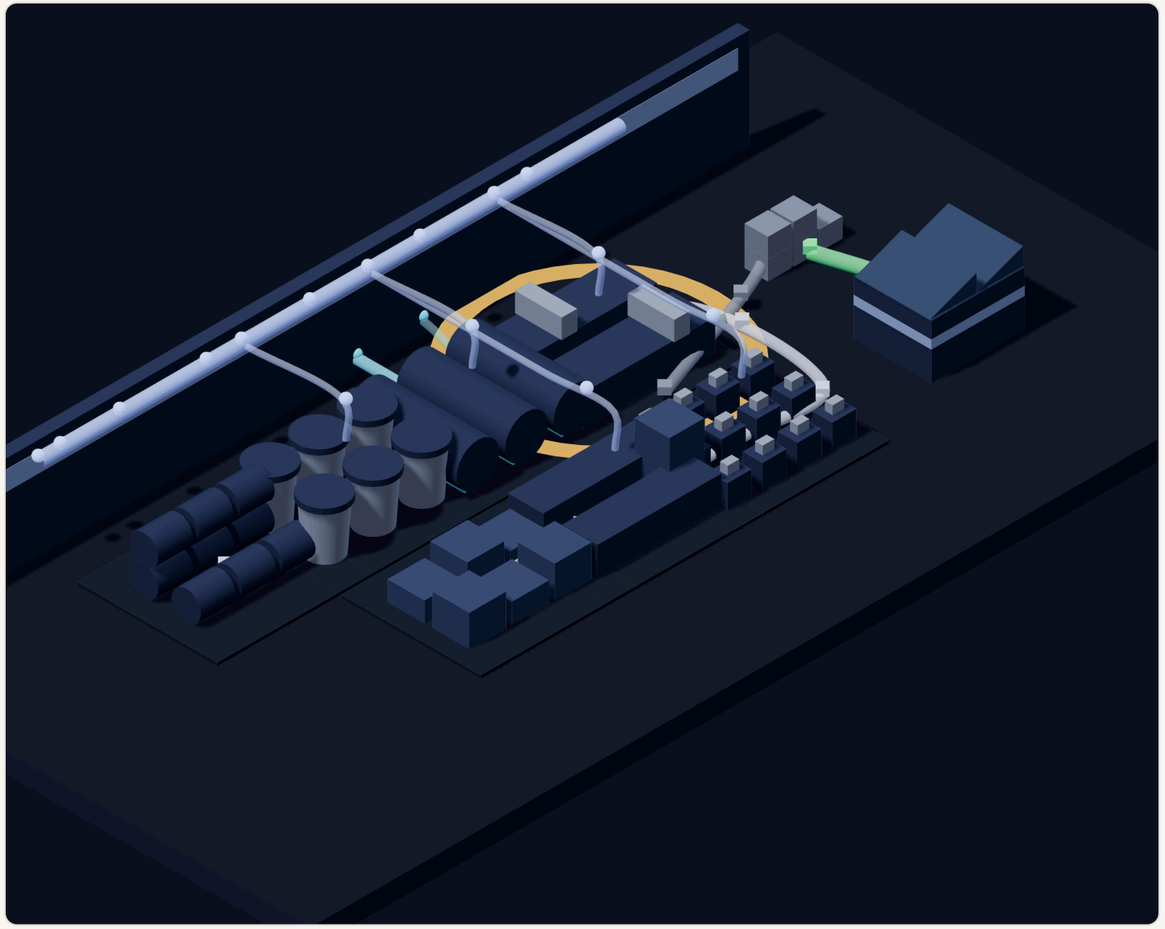
LEVEL 3 · HOW THE FACTORY OPERATES

Generative Factory Twin

Cutaway floor, seven process zones, energy bus with drops, water into dyeing, offcuts leaving Cutting along the green recovery path.



Light · Cutting selected

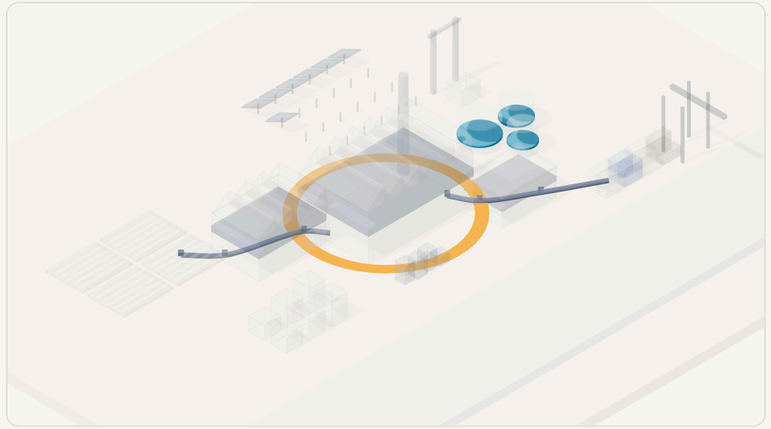


Dark

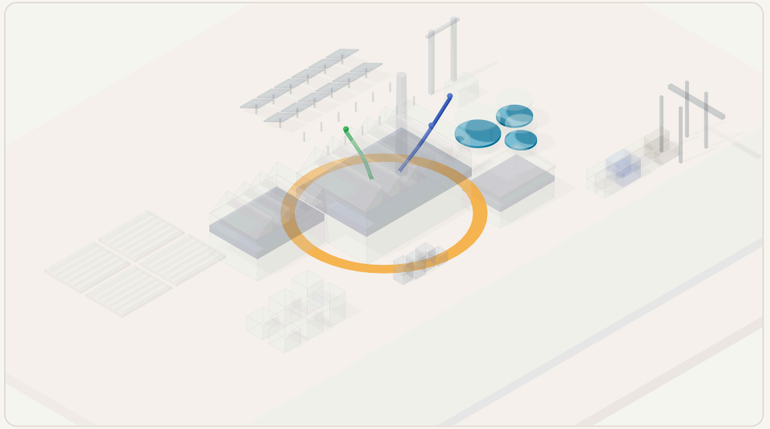
FLOW GRAMMAR

Material · energy · water · waste · circularity

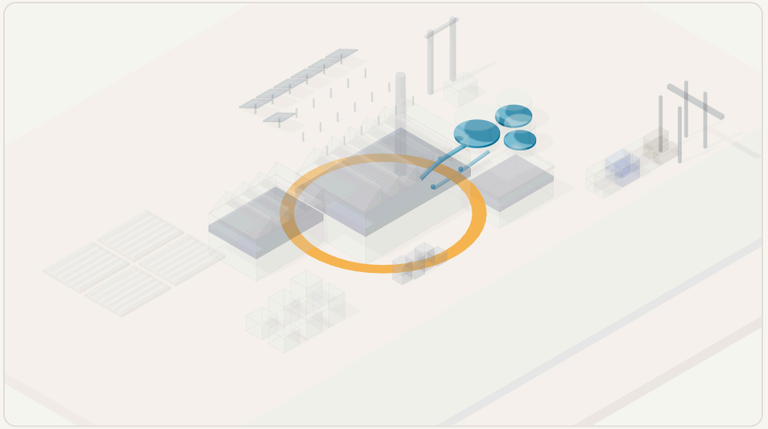
Each flow is typed by packet shape and token, so it reads without colour. Layer isolation ghosts the buildings to 22%.



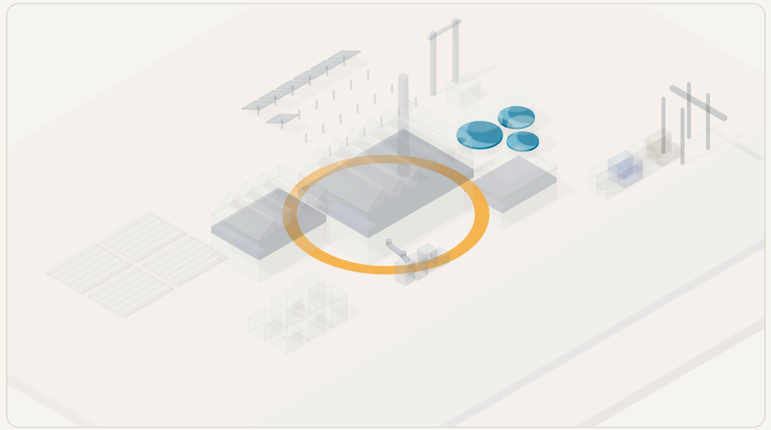
flow-material



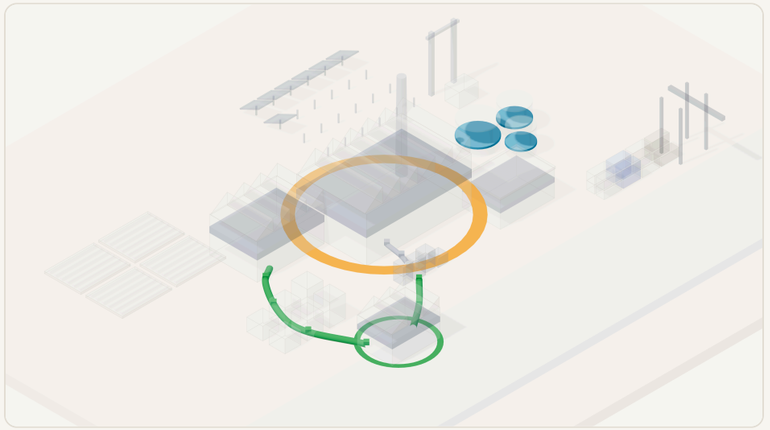
flow-energy



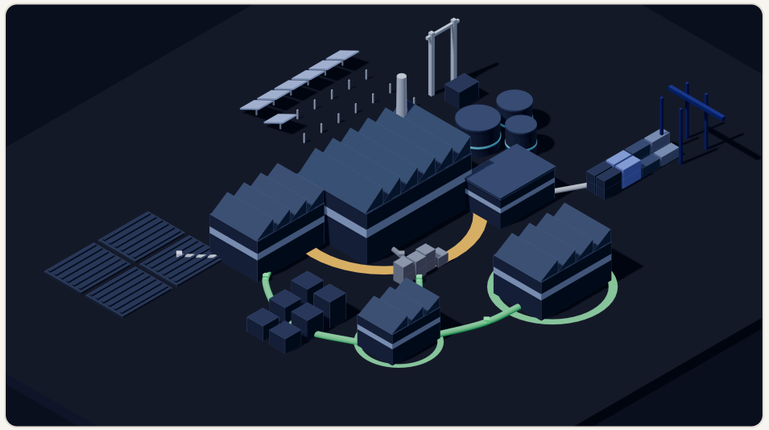
flow-water



flow-waste



flow-circular

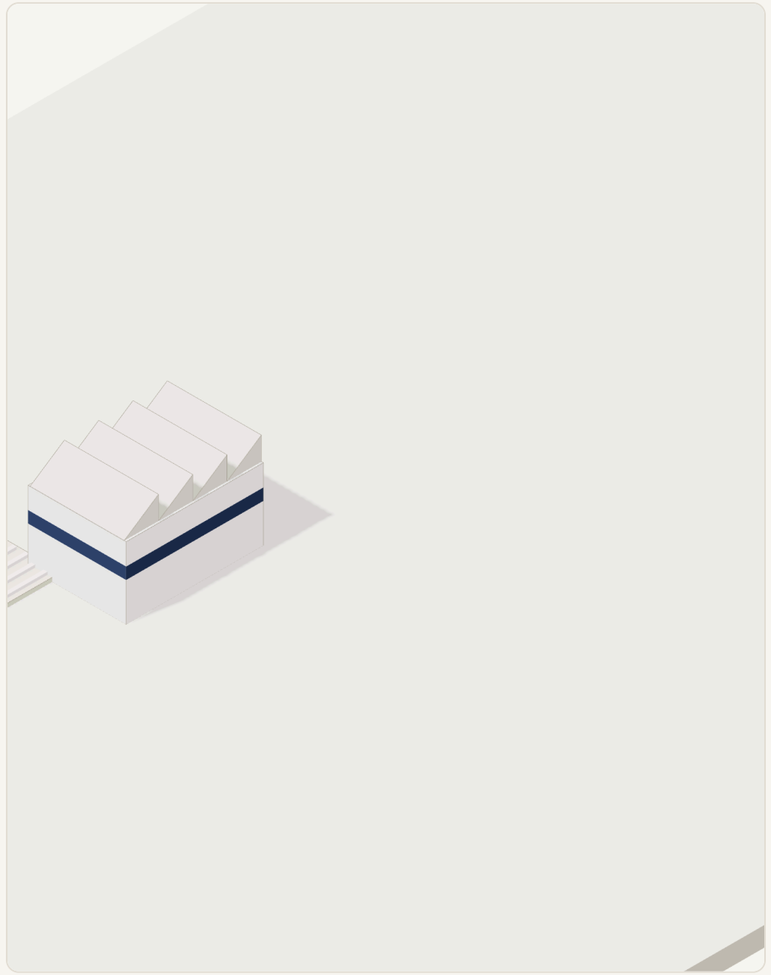


Factory A → waste → recovery → Factory B (expansion)

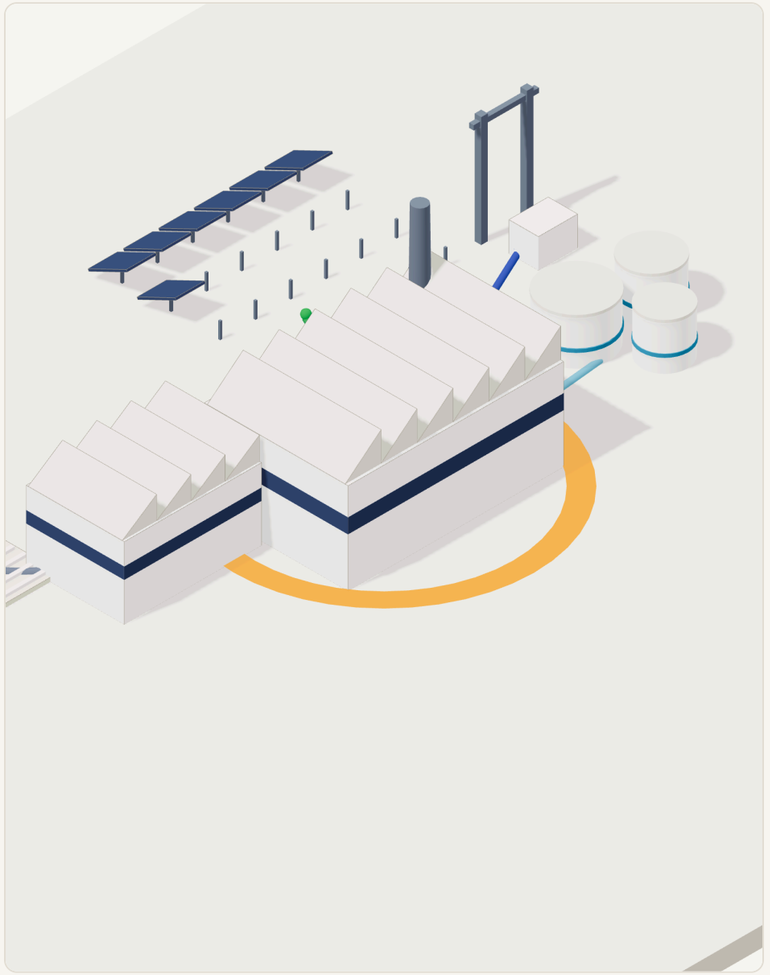
AI SCENARIO CONSTRUCTION

The ecosystem materialises step by step

Each workflow state drives exactly one scene change: supply → resources → complete.



1 · farms + spinning mill



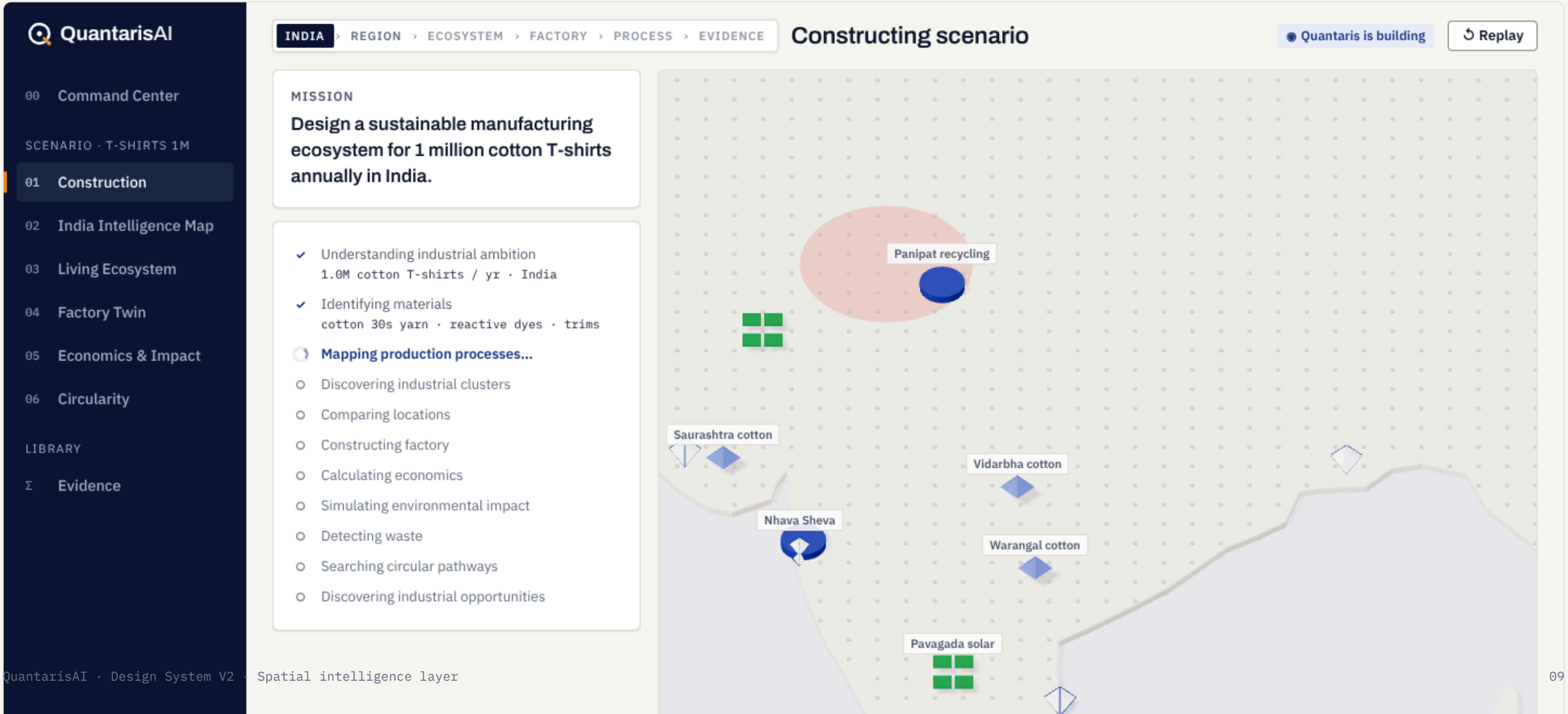
2 · factory, energy, water



3 · workforce, logistics, waste

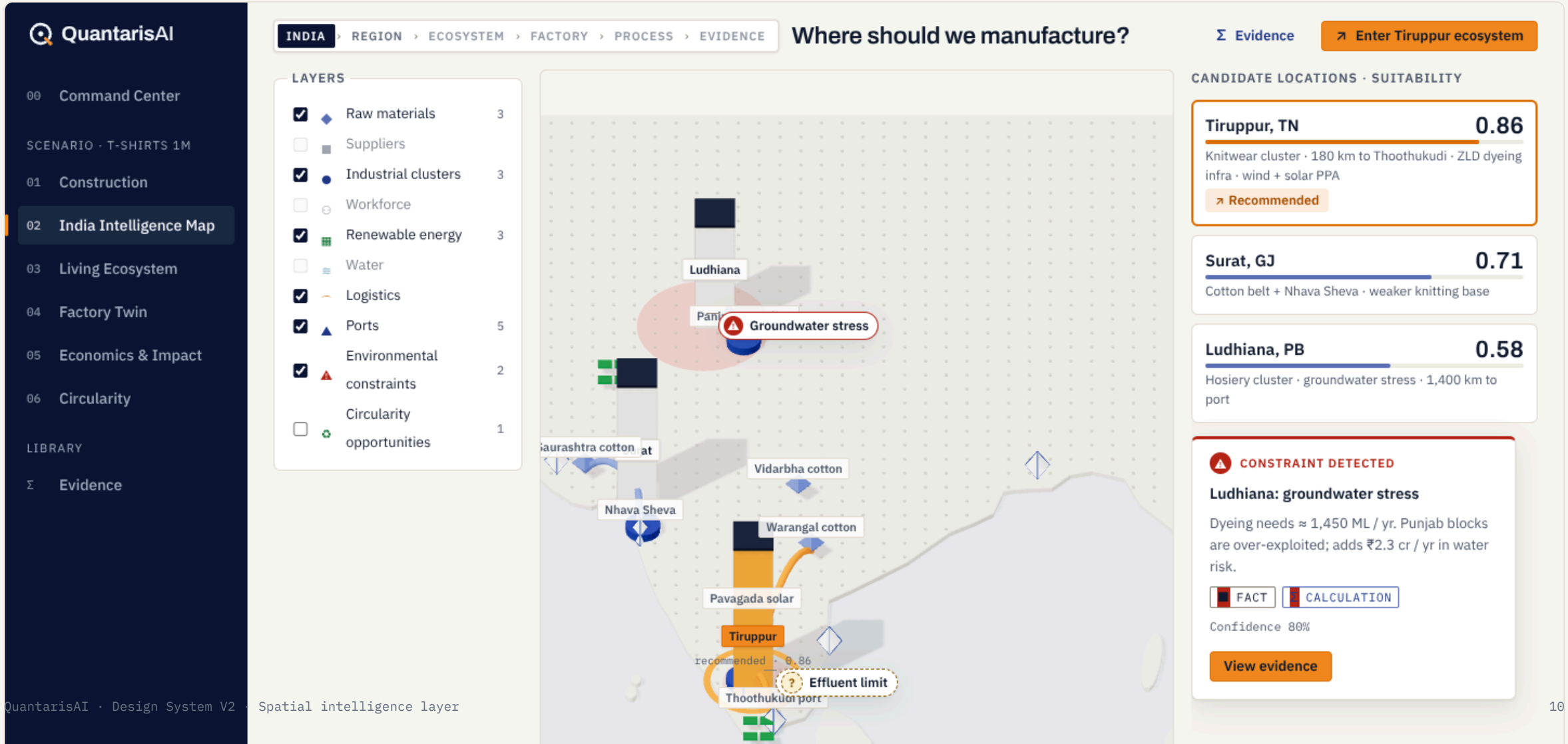
AI Scenario Construction

Construction feed + map → ecosystem materialising; CapacitySelector (A2UI) at step 2; proactive “New industrial opportunity” at the end.



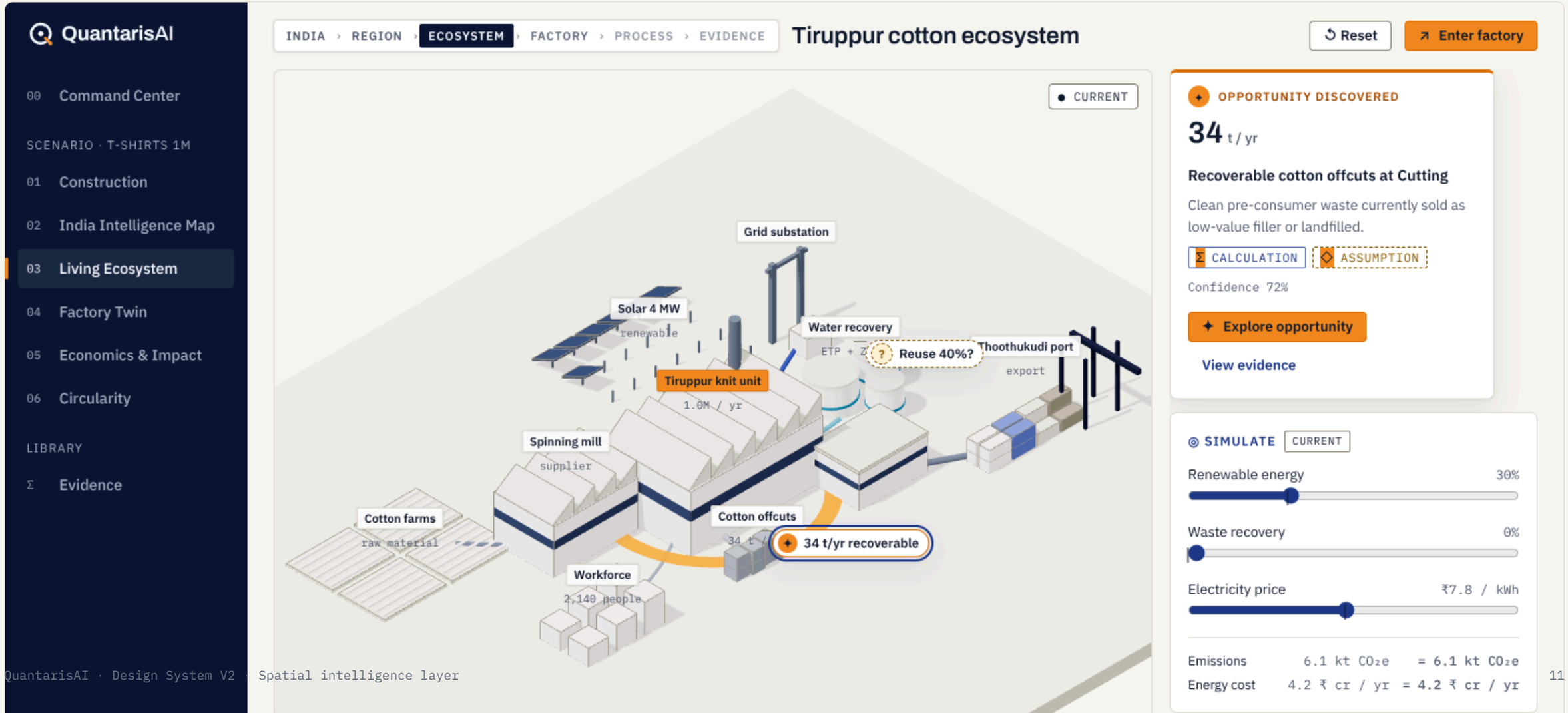
India Industrial Intelligence Map

Layers · 3D map · candidate comparison and constraint detection.



Living Industrial Ecosystem

Offcut marker → Explore → pathway search → green path → Model as new industrial opportunity. What-if simulation on the right.



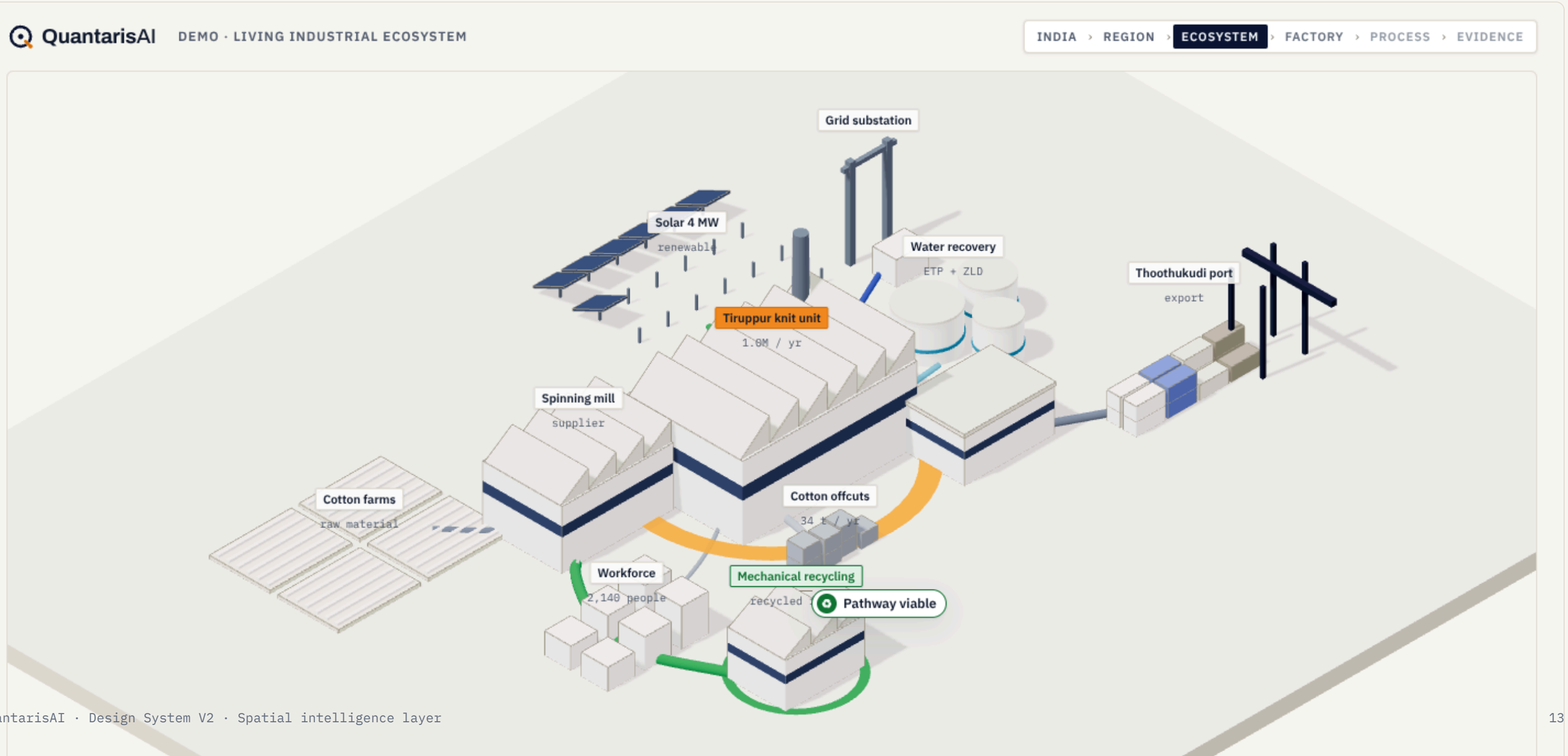
Generative Factory Twin

Select any process: capacity, utilisation, cost, emissions, waste, bottleneck. Evidence drawer opens over the twin.

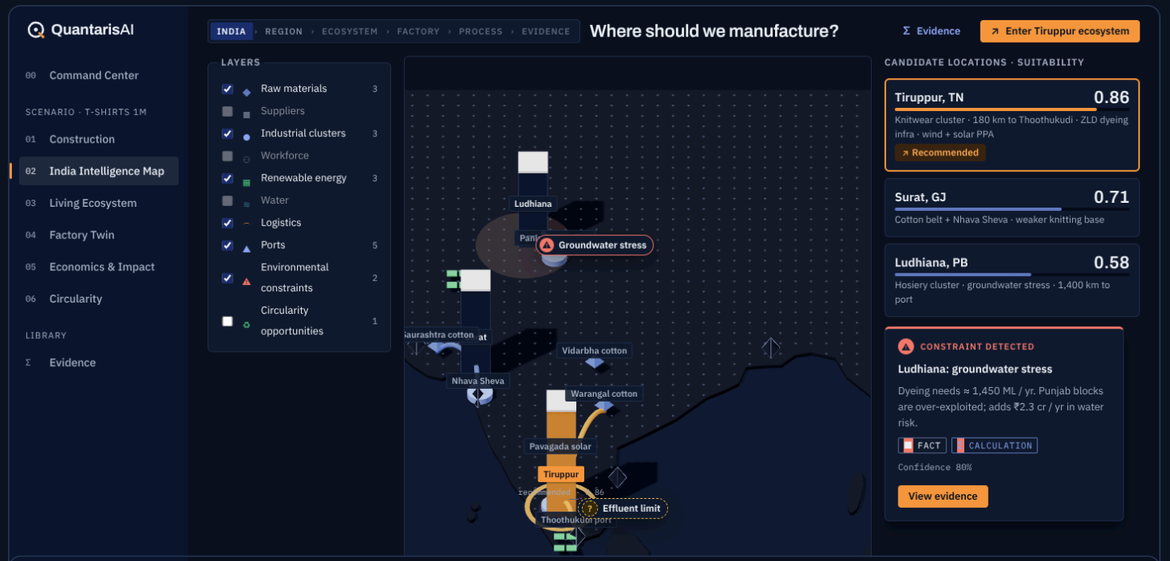


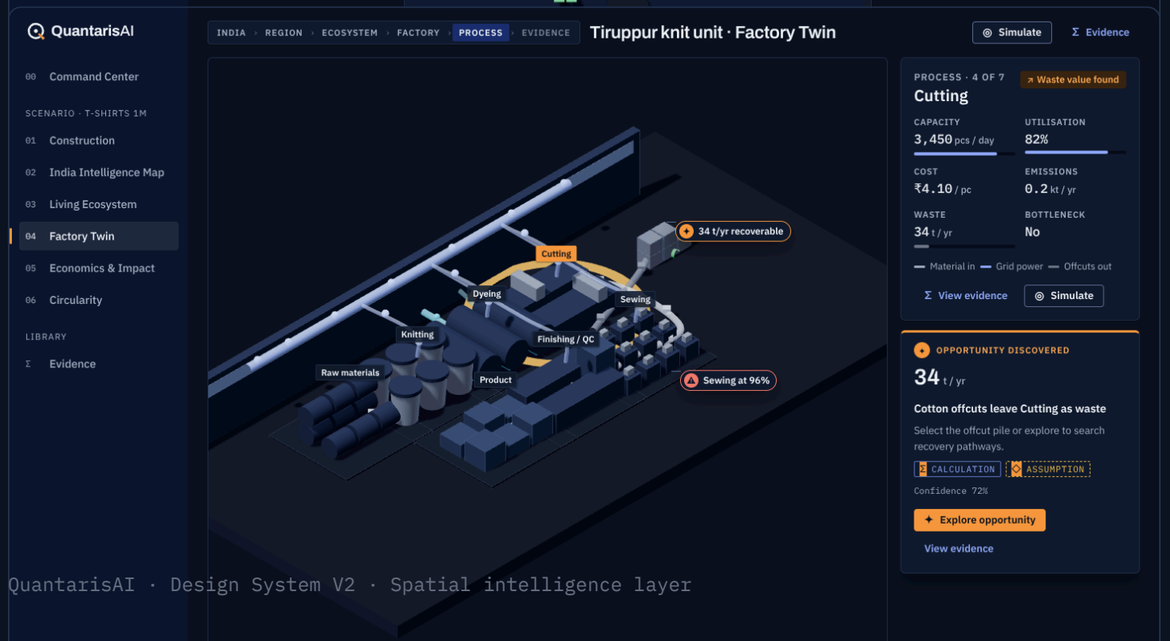
Hackathon demo walkthrough

The 14-beat demo story with Back / Autoplay / Next.



Deep navy, never black



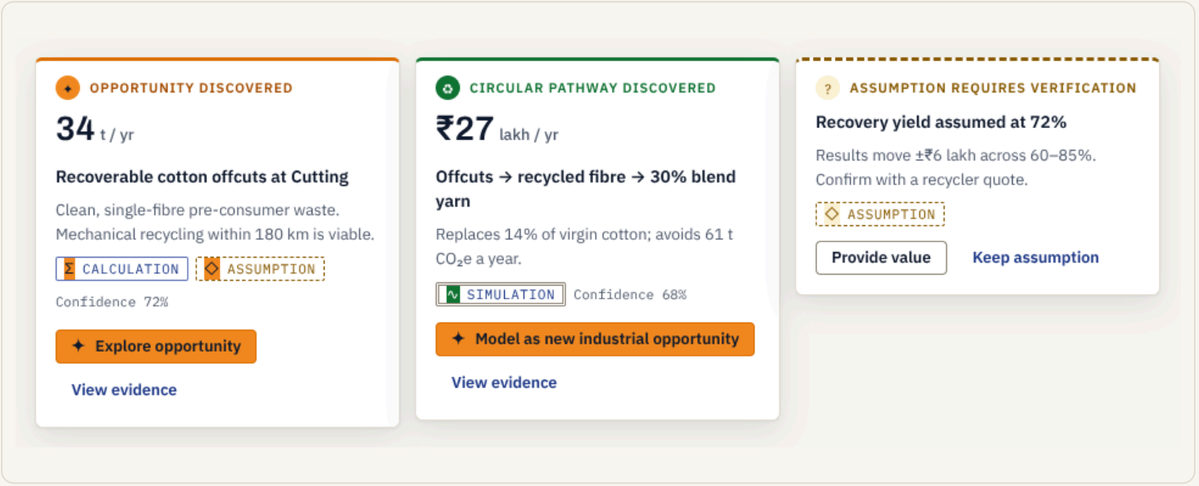




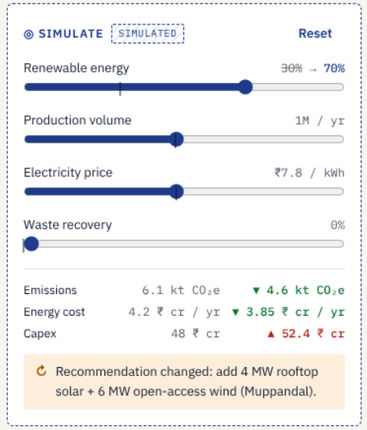
Living ecosystem

NEW COMPONENTS

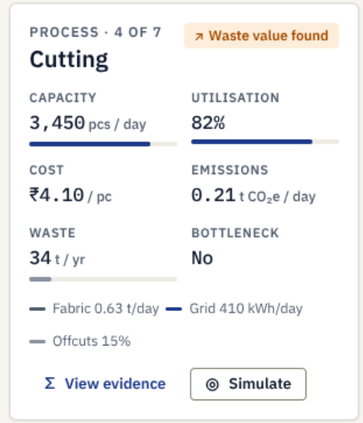
The AI layer and the simulation layer



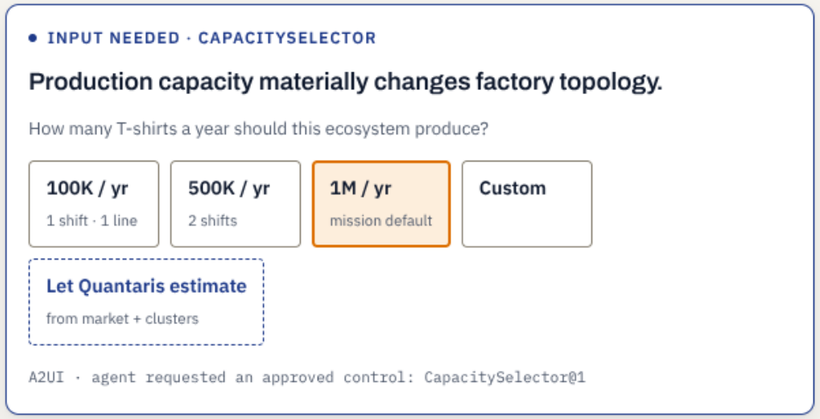
DiscoveryCard



SimulationPanel (SIMULATED)



ProcessInspector

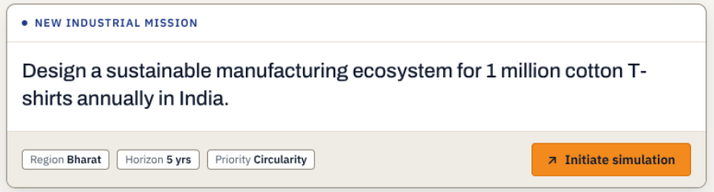


ClarifyCard · A2UI CapacitySelector



DiscoveryMarker · ✦ ⚠ ◎ ↗ ♻ ?

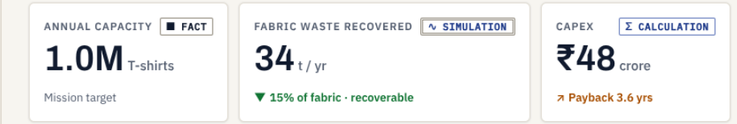
The enterprise information layer



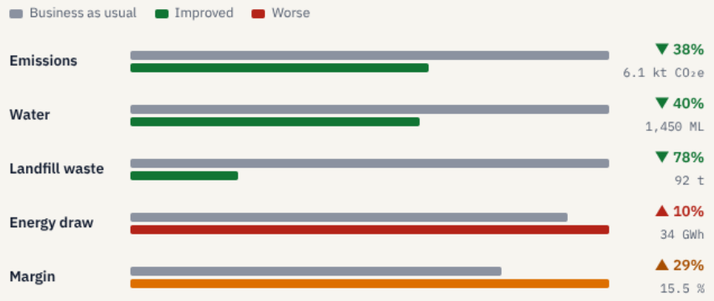
MissionComposer



StageRail



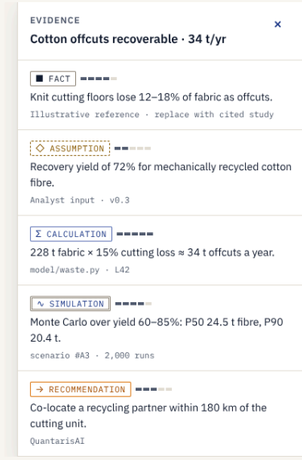
MetricTile



ImpactBars



CircularityPath



EvidenceDrawer

PACKAGE

Folder tree

```
QuantarisAI-V2-handoff/
├── AGENTS.md   # Codex instructions — read first
├── QuantarisAI-Design-System-V2.pdf   # visual overview (17 pages)
├── FOLDER-TREE.txt
└── design-system/
    ├── assets/
    │   ├── Logos/   # 5 SVG marks
    │   │   ├── qai-app-icon.svg
    │   │   ├── qai-mark-mono.svg
    │   │   ├── qai-mark-reverse.svg
    │   │   ├── qai-mark-small.svg
    │   │   ├── qai-mark.svg
    │   │   └── README.md
    │   └── Scenes/   # 15 reference renders 3200×2000
    │       ├── construction-1-supply.png
    │       ├── construction-2-resources.png
    │       ├── construction-3-complete.png
    │       ├── ecosystem-expansion.png
    │       ├── ecosystem-scene-dark.png
    │       ├── ecosystem-scene.png
    │       ├── factory-twin-dark.png
    │       ├── factory-twin.png
    │       ├── flow-circular.png
    │       ├── flow-energy.png
    │       ├── flow-material.png
    │       ├── flow-waste.png
    │       ├── flow-water.png
    │       ├── india-intelligence-map-dark.png
    │       ├── india-intelligence-map.png
    │       └── README.md
    ├── components/   # 25 components + 5 screens + cover
    │   ├── AppNav/                  README.md + preview.html
    │   ├── Button/                  README.md + preview.html
    │   ├── ChakraRing/              README.md + preview.html
    │   ├── CircularityPath/         README.md + preview.html
    │   ├── ClarifyCard/             README.md + preview.html
    │   ├── ConstructionFeed/        README.md + preview.html
    │   ├── Cover/                   preview.html
    │   ├── DiscoveryCard/           README.md + preview.html
    │   ├── DiscoveryMarker/         README.md + preview.html
    │   ├── EvidenceDrawer/          README.md + preview.html
    │   ├── EvidenceTag/             README.md + preview.html
    │   ├── ImpactBars/              README.md + preview.html
    │   ├── LayerPanel/              README.md + preview.html
    │   ├── LevelPath/               README.md + preview.html
    │   ├── lib/   # three.js r128 + scene kit (reference)
    │   │   ├── quantaris-scene.js
    │   │   └── three.min.js
    │   ├── Logo/                    README.md + preview.html
    │   ├── MetricTile/              README.md + preview.html
    │   ├── MissionComposer/         README.md + preview.html
    │   ├── ProcessFlow/             README.md + preview.html
    │   ├── ProcessInspector/        README.md + preview.html
    │   ├── Scene3D/                 README.md + preview.html
    │   ├── SceneLabel/              README.md + preview.html
    │   ├── ScreenConstruction/      README.md + preview.html
    │   ├── ScreenDemoWalkthrough/   README.md + preview.html
    │   ├── ScreenEcosystem/         README.md + preview.html
    │   ├── ScreenFactoryTwin/       README.md + preview.html
    │   ├── ScreenIndiaMap/          README.md + preview.html
    │   ├── SimulationPanel/         README.md + preview.html
    │   ├── SiteMap/                 README.md + preview.html
    │   ├── StageRail/               README.md + preview.html
    │   ├── StateView/               README.md + preview.html
    │   ├── StatusTag/               README.md + preview.html
    │   ├── bundle.css   # component styles
    │   ├── bundle.js   # reference React components
    │   └── index.d.ts   # props contracts
    ├── handoff/   # V2 implementation handoff — START HERE
    │   ├── contracts/   # JSON Schema 2020-12
    │   │   ├── a2ui.schema.json
    │   │   ├── agui-events.schema.json
    │   │   └── scene-state.schema.json
    │   ├── fixtures/   # demo data — reproduces the screens
    │   │   ├── agui-demo-run.json
    │   │   ├── demo-walkthrough.json
    │   │   └── scenario-tshirt-1m.json
    │   ├── tokens/   # generated from tokens.json
    │   │   ├── motion.json
    │   │   ├── quantaris-tokens.css
    │   │   └── theme.json
    │   ├── 3d-asset-requirements.md   # renderer, materials, objects, budgets
    │   ├── components.md   # inventory + layer per component
    │   ├── implementation-plan.md   # stack, repo layout, milestones
    │   ├── interactions.md   # trigger → result → event
    │   ├── README.md   # Codex entry point + read order
    │   ├── responsive.md   # breakpoints, sheets, WebGL fallback
    │   └── screens.md   # 5 screens · STATIC / R3F / MAPS / AI per region
    ├── ai-native-interaction.md   # construction, discovery, signature moment, A2UI
    ├── data-and-evidence.md   # charts + evidence language
    ├── design-system.json   # index: libraries, asset groups
    ├── engineering-handoff.md   # summary → handoff/
    ├── motion.md   # motion spec
    ├── product-surfaces.md   # V1 surface rules
    ├── README.md   # brand book
    ├── spatial-intelligence.md   # 3 levels, scene + flow grammar
    └── tokens.json   # design tokens — source of truth
```